Bayesian Hierarchical Models for Expected Goals, Temporal Stability, and Possession Value in the AHL

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This study presents Bayesian approaches to quantify how expected goals vary by shot location and player, whether that relationship is stable across a season, and how individual puck actions change the probability of generating a dangerous offensive outcome within 10 seconds given spatial and game context. The first model, a Bayesian hierarchical logistic regression, confirmed that shot location dominates over individual finishing skill. The second model formally tested the temporal stability of the xG surface across 10-game windows, confirming the static xG surface as a reliable seasonal summary. The third model revealed that puck action context substantially outweighs location in predicting danger, with one-timers, successful actions, and slot passes as the dominant predictors. Player skill in generating dangerous situations proved substantial, a finding invisible to traditional location-only xG models.

1. INTRODUCTION

In this section, I introduce background information, past literature, and our proposed research question.

1.1. Background

Hockey analytics has lagged behind other sports like baseball in the boom of statistical modeling. In the years 2015-2020, hockey analytics exploded, with significant investment made in the hockey industry. Companies like Sportlogiq introduced detailed event and tracking data. Alongside league investment, the NHL began rolling out player and puck-tracking systems. The NHL teams then started dedicating departments to analytics. By 2018–2020, almost every NHL team had an analytics staff.

At the 2012 SLOAN Conference, the first Expected Goals (xG) model of hockey was presented. A method for team and player evaluations that started off as shot quality models, then evolved to more specifically "what is the probability that any given shot will result in a goal?" With lots of ways to answer this question now.

As of 2023, Wisehockey introduced the world's first automated, real-time EPV statistics for ice hockey. Expected Possession value for ice hockey measures how likely it is that a team's puck control leads to scoring within the following eight seconds. Where: High EPV indicates players who consistently progress play, mitigate danger, or create high-value plays from low-value situations.

Traditional hockey evaluations have relied on goals and assists. These are poor metrics as they are rare and random events, thus not a good measurement of player or team performance. Overall, a challenge in hockey is the sparsity of its player-level data. The raw shooting percentage of a player, when players may only take 30-80 shots in an entire season, is an unreliable estimate of a player's true ability. Partial pooling in Bayesian hierarchical models addresses this issue by borrowing information across players to shrink estimates for players with low action volume toward the team's mean. This allows high-action-volume players to deviate meaningfully from it. This property makes Bayesian methods particularly well-suited to hockey, where sample sizes vary dramatically across players, and any single-season estimate carries substantial uncertainty.

1.2. Literature Review

A Player and Position Correction for Expected Goals (xG) using a Bayesian Hierarchical Approach has been applied to English Premier League data. In this study, they developed several Bayesian models to assess the influence of player-level effects on xG predictions. It was found that when additional predictors were added to the model, the player positional effects became close to insignificant, "suggesting that player position had minimal impact on xG when considering more shot-related factors".[1]

Mentioned briefly above, the Expected Goals (xG) model for Evaluating NHL Teams and Players set the stage for xG models. The study used many predictors to measure the positive or negative impact a player had on their team while on the ice. As in a positive plus-minus, if they had a positive effect on their team's expected goals when on the ice. A key variable seemed to be hits, with teams that allow more hits against them tending to score more goals. Typically, the team doing the hitting does not have the puck. Therefore, good puck possession teams simply have the puck more often. Overall, xG has since been found to be a better predictor of future performance than actual goals scored.

Past research emphasizes the importance of expected-possession-value models for soccer analysis and, therefore, suggests they could be a beneficial approach for hockey. Soccer → Expected Threat (xT) → Hockey Possession Value Model Hudl is a highly reputable and industry-standard source for athletic performance analysis. The possession value framework is defined as being able to objectively and quantifiably measure the value of each event on a soccer pitch. Expected Threat (xT) is calculated by laying a 'value surface' over a football pitch to divide it into zones, where each zone has a value assigned to it based on how likely a goal is to be scored from that zone [2]. This framework was introduced in the industry over the last decade. It is important to note that the model only values actions that move the ball from zone to zone. For example, passes and carries would be in the model, but not defensive moments or shots. Possession Value uses a time-based approach, measuring the probability that a team in possession will score during the next 10 seconds of the play. So, it measures how much a pass or carry increases the probability of scoring in the future, not immediately.

This was the introduction of the Markov game framework to hockey analytics. The approach involves context and lookahead variables across play-by-play sequences.[3] Player actions impact ranged greatly depending on the context of the play. A limitation was that shot location was not incorporated, which the following study has included.

A Markov game model for valuing actions, locations, and team performance in ice hockey incorporates the location of an action on the ice into the value function. The same action, being a pass, a dump-in, or a carry, has meaningfully different values depending on where it happens on the ice.[4] This study is the first to quantify the difference in worth across the same action but at different locations.

Bayesian space-time models for expected possession added-value study reveal that offensive sequences develop through Markov decision processes, modeling the flow of event buildup within a sequence. On the ice rink, spatiotemporal transition probabilities are mapped. Finally, determining the Expected Possession Value (xPV) by running simulations with the Possession Added Value (PAV) calculated as the sequence progresses.

1.3. Research Questions

Hockey is a fast-paced sport where players must make the best decision in a short time. Players must work together to create a chain of plays to manufacture as many scoring chances as possible. The goal of this study is to quantify how xG changes given different variables. Along with how that plays into different types of puck actions meaningfully increases the probability of generating a dangerous offensive outcome in the next 10 seconds.

1. How does xG (Expected goals) vary by shot location and player?

2. Is that xG relationship stable across time, or does a player's shooting probability from a given location drift? Specifically across a 10-game window, or is part 1 reliable?

3. How does each puck action change the probability of generating a dangerous offensive outcome within the next 10 seconds, given spatial and game context?

This is interesting to coaches, general managers, and analysts because the research provides quantifiable evidence on how individual puck actions and shot locations drive scoring probabilities. This study transforms subjective hockey intuition into measurable and quantifiable intelligence. Understanding which actions most reliably lead to danger allows teams to design systems and evaluate players on the quality of their decisions, not just on the goals that result. At the player valuation level, separating location-driven xG from genuine finishing skill gives front offices a principled framework for distinguishing a shooter who scores because he finds the right spots from one who converts chances others would miss.

2. METHODS

In this section, the tools, data cleaning and transformation, model architecture, model prior assumptions, and model validation are discussed.

2.1. Tools

This study was conducted using R 4.4.3. The packages I used are tidyverse, arrow, ggplot2, dplyr, brms, tidybayes, loo, patchwork, and grrepel.

2.2. Data Cleaning and Transformation

The datasets I will analyze were shared with me directly by the Colorado Avalanche analytics team as part of the HALO Hackathon. The Colorado Avalanche obtains its data through the NHL Edge, a high-tech system involving active infrared sensors embedded in pucks and sewn into players' jerseys. Thus, the data consisted of event-level observations from 480 AHL games, with tracking-enhanced data. The data is observational and was given across five different datasets: games, events, stints, players, and tracking.

The first step in data preparation is for building the goals prediction model. For the event type in the events data, only shots and goals are kept, and other events, like passes or fouls, are excluded. Then, dropping missing coordinates in the x or y, outcome, and player ID variables. Next, creating a binary variable from the goal event type to (1) a goal occurred from the shot, (0) a goal did not occur from the shot. Lastly, standardizing coordinates to have a mean of 0 and a standard deviation of 1. Then I ran a check to confirm the spread of goals was correct, since about 5% of shots were goals, which is consistent with the average for an NHL team.

For the second research question, longitudinal shot data are needed to assess whether the calculated static xG surface is stable across a season, specifically, whether it drifts over a 10-game window. The key estimate is sigma drift, which is the standard deviation of how much a player's xG-location relationship shifts over time . First, building the longitudinal data, starting with the cleaned shots data from question one. I join the game dates from the games dataframe to order the shots chronologically, ranking each player's games so a player's first game is a rank of one, and so on. Then, each shot is assigned to a 10-game time block (window), filtering to players with at least two game blocks to keep the model consistent. I also rescaled the coordinates using the same mean and standard deviation as in question one, ensuring the spatial scale is identical across both parts so the results are directly comparable. The next section aggregates the data to the player-time-block level, collapsing it into one row per player per window. Each row records goals, shots, average x- and y-shot locations, and previous block and block-sequence tracking, with block occurrences ordered. This aggregation makes the temporal model tractable because, instead of 56,000 individual shot rows, I have about 1,700 player-block summaries that directly capture how each player's goal rate evolves across the season. This is done by creating a lagged block for the AR(1) structure, using the previous block's index. Lastly, a simple quality check to see how many blocks with fewer than five shots produce very noisy goal rates. The result showed only 73 of 1,716 blocks (4%) fall below this threshold, confirming the aggregated data is reliable.

For the third research question, it answers how each puck action changes the probability of generating a dangerous offensive outcome, within the next 10 seconds, given spatial and game context. First, I defined what a dangerous outcome is and built an outcome window. I sorted each event within each game and created two new columns. One being absolute time, which converts period and time into a single absolute second count within the game. This is necessary because I need to measure time gaps between events across period boundaries. The objective is to look ahead 10 seconds for each event and flag if a shot or goal occurs. I also created a variable to find the next shot or goal for each event within the 10-second window. For each i, it grabs at most the next 20 events and checks the conditions. It must be a shot or a goal, it must be on the same team, and it must have happened within 10 seconds. All three must be true for the dangerous next to be flagged as true. The result is a binary TRUE/FALSE label on every event, which becomes my model's outcome variable. Next removes events without spatial or player attributes. I also removed any non-puck actions such as faceoffs, whistles, and period endings. This is because they are not actions a player chooses to take. Then the outcome bin is the dangerous outcome, which is converted to a binary (1 or 0) for the Bernoulli model. Is successful is a new variable indicating whether the action was deemed successful; a successful carry differs from an unsuccessful one. I also normalized the spatial coordinates, which flips all events to the same offensive grid. The same as part one and part two. The nature of the actions is extracted into binary features from the flag column. Such as a one-timer attempt, seam, and low-high. Then, event type-specific contexts are extracted from the detail column. Lastly, I kept the eight most common event types and collapsed the remaining event types into an Other category. This sets pass as the reference category, so all other action coefficients are interpreted relative to a pass. Pass is the natural baseline as it is the most common action and has a 26.5% danger rate, right in the middle of the spectrum of different actions. This is confirmed by a danger-rate table by action type, created to validate the feature engineering before modeling. Reception (33%) and carry (31%) at the top, which makes intuitive sense. Receiving a pass means you just got the puck in position to shoot, and carrying the puck means you're driving toward the net. Block (13%) and failed pass location (11%) at the bottom also make sense. These are either defensive actions or turnovers where the team loses possession, and danger drops immediately.

2.3. Assumptions

Below, I list the priors I used to answer the research question one. For the location fixed effects, I set the prior to Normal(0, 1.5). Zero at the center to make the data prove the importance of the location variables in any direction. A standard deviation of 1.5 is set to allow more freedom of range for the coefficients, as large location effects are possible but will not be assumed. The average shooting percentage is roughly 9-10% for shots on goal, and slightly higher, 10-12%, when you include all shots attempted [5]. So the empirically observed 9-12% shooting range maps to a log-odds of about -2.0 for the center with a standard deviation of 1. The Bayes-xG paper found that player-level effects persist even after adding location predictors, which will directly inform the prior choice for the player random effects [6]. Thus, choosing Exponential(1) with the reasoning of the player intercept standard deviation is meaningfully above but shrinking towards zero. After I tried Exponential(1) for the player-to-player variation, I changed my prior to Exponential(2) as there were very small player differences in the model; thus, by switching over, I made my prior more accurate.

Below, I will list the priors I used to answer research question number two. The same prior, Normal(0, 1), is used for the fixed effects for shot location because it has worked well. The between-player baseline differences had a prior of Exponential(2), also the same as part 1. The prior intercept is Normal(-3, 1), centered at -3 on the log-odds scale, corresponding to a baseline block-level goal rate of around 4.5%, which is directly consistent with the observed mean in the data of 1.51 goals per 30.3 shots. Players in the AHL are shooting at rates closer to 7-8% [7]. Therefore, the block-level aggregation brought it closer to about 5% because it averages across all player types. Converting 5% to log-odds yields about -2.94, which justifies the -3 center. The standard deviation of one is loose for weakly informative priors, regularizing population variances while still allowing the data to move estimates away from the prior center. Lastly, the Exponential of rate two for the time block encodes prior skepticism that player shooting profiles drift substantially across 10-game windows. The choice of rate equals 2 sets a prior mean of 0.5 on the log-odds scale. This is lenient enough to detect real drift if it exists, but concentrates most of the mass near zero to reflect the domain expectation that shooting behavior is largely stable within a season [8].

Below, I list the priors I used to answer research question three. The intercept prior Normal(-1, 1) is centered at about 27% baseline risk probability, directly calibrated to the observed rate of 21.9% in the data (log(0.219/0.781) =- 1.27), with a standard deviation of one [8]. The global fixed effect prior Normal(0, 1.5) is made wider than parts one and two, reflecting that action type effects could be substantially larger than pure location effects, which is consistent with papers [9]. The Expected Threat framework demonstrated that action type explains more variance in scoring probability than spatial location alone. One-timer, seam pass, and action success priors use Normal(0, 1.5), which is meaningful but not excessive, based on [10], who found that transition actions had substantially higher scoring values than static zone possessions. Odd-man rushes, slot shots, and slot passes receive the wider Normal(0, 2) distribution, with [11] found slot and rush situations generating disproportionately high possession value across all other action-location combinations, justifying a prior permitting larger coefficients. The exponential of the rate two is on the player random intercept standard deviation, which follows a principled complexity penalty prior [12], encoding skepticism that large between-player differences persist after controlling for action type, location, and all contextual features.

2.4. Model Architecture

Question 1: How does xG (Expected goals) vary by shot location and player?

To address research question one, I built a predictive model for xG using the `bf` formula from the `brms` package to assess its support for random effects. The model is set up to include fixed-location effects for `x_scaled` and `y_scaled` to capture the population-level shooting probability. Along with scaled quadratic location terms to let the xG surface curve. Then, after controlling for location, we have a partial pooling term so that each player gets their own intercept. Next, priors are set based on the assumptions discussed in the previous section. Then fitting the model with `brm` from Stan. To fit the model, I used the Bernoulli family for the binary outcome, running 2 chains with 1000 iterations, 4 cores, and `adapt_delta=0.95` in the control to reduce divergent transitions, which are common in hierarchical models.

Then the model checks were run, including Rhat, trace plots, and posterior predictive checks. Rhat of 1 or 1.01 displayed that all chains agree and converge. Effective sample size (ESS) for parameters is roughly 600-1200, except for player ID. The lower ESS for the player random effect is expected, as true player-to-player variance is near zero, confirmed by the sd(Intercept) estimate of 0.05. This is true for hierarchical variance parameters and does not invalidate the model. The Rhat of 1.00 confirms the chains have converged to the same distribution. Trace plots show "fuzzy caterpillars," which means no sign of drifting. The posterior predictive check simulation overlay showed that the model-generated goal rates match the actual rate of about 5

```
 Family: bernoulli
  Links: mu = logit
Formula: goal ~ x_scaled + y_scaled + I(x_scaled^2) + I(y_scaled^2) + (1 | player_id)
   Data: shots (Number of observations: 56787)
  Draws: 2 chains, each with iter = 1000; warmup = 500; thin = 1;
         total post-warmup draws = 1000

Multilevel Hyperparameters:
~player_id (Number of levels: 1044)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.04      0.03     0.00     0.12 1.01      301      494

Regression Coefficients:
            Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept      -2.51      0.04    -2.59    -2.44 1.01     1535      576
x_scaled        0.69      0.03     0.63     0.75 1.01     1090      667
y_scaled        0.06      0.03    -0.00     0.13 1.00     1377      643
Ix_scaledE2    -0.11      0.03    -0.16    -0.05 1.00     1004      918
Iy_scaledE2    -0.96      0.04    -1.04    -0.87 1.00     1073      833

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Question 2: Is that xG relationship stable across time, or does a player's shooting probability from a given location drift?

To address research question two, I built a binomial mixed model that models goal success from shot attempts, with fixed effects controlling for shot location, connecting back to part one. Along with moving from a random intercept for each player in part one to a random slope on the 10 window time block. The steepness of each player's line is their individual slope. Some players may improve across windows, some decline, and most stay flat. The standard deviation of all those slopes across every player is the sigma drift variable. Next, priors are set according to the assumptions discussed in the previous section. Then, fitting the model with brm, the binomial family is used for goals out of trials of shots, with 4 chains run for 3000 iterations each, using 4 cores, and set adapt_delta to 0.97 for the control to reduce divergent transitions, which are common in hierarchical models. Even higher than in part one because this model has hierarchical slopes that need more fine-tuning. Along with a max tree depth of 12, to prevent the sampler from getting lost in complex geometry.

Then, quick diagnostic testing was completed. All Rhat's are 1.00 with all the variables ESS in the thousands. All fixed effects show perfect hairy caterpillar mixing across all 4 chains, all clean. Both standard deviation parameters show the expected near-zero funnel behavior, but Rhat = 1.00 confirms convergence despite the spikiness. This is the same pattern seen in part one and is acceptable given that the true variance is near zero. The correlation trace plot is nearly flat across the entire negative one-to-one range, and the plot shows all four chains jumping randomly with no structure. This confirms why using —— in the formula is the right move. When both the intercept and slope variance are near zero, their correlation is mathematically indeterminable. Lastly, the posterior predictive check showed that the model closely tracks the data throughout, confirming that it correctly reproduces the overall distribution of goal counts. The wavy oscillating pattern in both the observed and predicted lines reflects the discrete nature of goal counts per block. Most blocks have 0, 1, 2, or 3 goals, creating alternating peaks and troughs in the density. The model reproduces this discreteness, emphasizing that binomial likelihood is a well-specified choice for this data structure.

```
 Family: binomial
  Links: mu = logit
Formula: goals | trials(shots) ~ mean_x + mean_y + (time_block | player_id)
   Data: player_blocks (Number of observations: 1716)
  Draws: 4 chains, each with iter = 3000; warmup = 1500; thin = 1;
         total post-warmup draws = 6000

Multilevel Hyperparameters:
~player_id (Number of levels: 671)
                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)                 0.06      0.04     0.00     0.16 1.00     1815     2643
sd(time_block)                0.03      0.02     0.00     0.08 1.00     1662     2575
cor(Intercept,time_block)    -0.12      0.46    -0.88     0.76 1.00     4082     3522

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    -3.09      0.02    -3.14    -3.05 1.00     7414     4836
mean_x        1.03      0.05     0.94     1.12 1.00     7870     4583
mean_y        0.12      0.06    -0.00     0.25 1.00    11139     4438

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Question 3. How does each puck action change the probability of generating a dangerous offensive outcome (e.g., shot or slot shot) within the next 10 seconds, given spatial and game context?

To address research question three, I built a Bayesian logistic regression model that predicts whether a given puck action leads to a dangerous outcome within the next 10 seconds. The formula has many predictors. The key variable is action type, with pass as the reference level. Each coefficient answers that relative to a pass, how much does this specific action type change the log-odds of generating danger in the next 10 seconds?. The same spatial fixed effects for x scaled and y scaled as part one. The contextual enrichment features that go beyond location are: Does outnumbering defenders increase danger? Does a matched rush increase danger vs static play? Is shooting from the slot more dangerous than other shots? Does passing to the slot increase danger more than other passes? Do passes off the rush create more danger? Does entering the offensive zone create immediate danger? Do one-timers substantially increase danger? A one-timer is defined as a shot taken in one motion, without stopping the puck first. Do seam passes through the middle increase danger? Does action success matter beyond action type? Also, a period variable controls for game state, as teams play differently across periods 1-3. After controlling for everything above, some players consistently pose a greater danger than others, regardless of the action they take or where they are. A partial pooling term captures that individual skill component. Lastly, 20% of the modeling data is subsampled. The full model data has over 1 million rows. The 20% subsample gives about 200,000 observations, which is still a big enough sample for estimating stable coefficients. The outcome rate remains at 0.219, so the subsample is still representative of the entire dataset. Then, fitting the model with brm, it used the Bernoulli family for the binary outcome, ran 2 chains for 1500 iterations each, 4 cores, and adapt_delta of 0.95 in the control to reduce divergent transitions, which again are common in hierarchical models. Along with a max tree depth of 12 to prevent the sampler from getting lost in complex geometry.

```
 Family: bernoulli
  Links: mu = logit
Formula: outcome_bin ~ action_type + x_scaled + y_scaled + I(x_scaled^2) + I(y_scaled^2) + is_odd_man + is_even_rush + is_slot_shot + is_slot_pass + is_rush_pass
+ is_zone_entry + is_timer + is_seam + is_successful + period + (1 | player_id)
   Data: model_data_small (Number of observations: 150753)
  Draws: 2 chains, each with iter = 1000; warmup = 500; thin = 1;
         total post-warmup draws = 1000

Multilevel Hyperparameters:
~player_id (Number of levels: 1170)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.37      0.01     0.34     0.39 1.00      244      596

Regression Coefficients:
                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept                -1.89      0.03    -1.95    -1.84 1.00      800      859
action_typeblock         -1.24      0.25    -1.73    -0.77 1.00     1695      643
action_typecarry          0.55      0.03     0.49     0.60 1.00     1287      735
action_typelpr           -0.36      0.02    -0.39    -0.32 1.00     1165      882
action_typereception      0.64      0.04     0.56     0.72 1.00     1599      939
action_typeshot          -0.08      0.03    -0.14    -0.01 1.00      735      681
action_typeOther         -0.90      0.04    -0.98    -0.82 1.00     1614      802
x_scaled                 -0.01      0.01    -0.03     0.00 1.00     1362      898
y_scaled                  0.00      0.01    -0.01     0.01 1.00     2025      671
Ix_scaledE2              -0.32      0.01    -0.34    -0.30 1.00     1448      853
Iy_scaledE2              -0.08      0.01    -0.09    -0.06 1.00     1286      829
is_odd_manTRUE            0.04      1.97    -3.66     3.99 1.00     1299      673
is_even_rushTRUE         -0.01      1.46    -3.02     2.82 1.00     1464      825
is_slot_shotTRUE          0.13      0.05     0.03     0.22 1.00      929      825
is_slot_passTRUE          1.08      0.04     1.00     1.15 1.01     1054      736
is_rush_passTRUE          0.67      0.05     0.58     0.77 1.00     1386      651
is_zone_entryTRUE         0.14      0.04     0.06     0.23 1.01     1275      822
is_timerTRUE              1.54      0.04     1.45     1.63 1.01     1202      512
is_seamTRUE               0.34      0.05     0.25     0.43 1.00     1229      828
is_successful             1.33      0.02     1.29     1.36 1.00     1562      792
period                   -0.00      0.01    -0.02     0.01 1.00     1991      569

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Then, quick diagnostic testing was completed. All Rhat's are 1.00 across every parameter, perfect convergence even with only 2 chains. EES values are all comfortably above 500, with most being above 1000. The only ESS below is for the player-variance sampling, but it still converges to an Rhat of 1.00. Key findings were derived from action-type coefficients relative to passing as the reference. The reception estimate is 0.64, meaning receiving a pass is the most dangerous action type. Next, carrying has an estimate of 0.55, making it the next most dangerous. Loose puck recovery reduces danger by an estimated -0.36. Blocking strongly reduces danger, with an estimate of -1.24. For contextual features, is_timer is the strongest predictor with an estimate of 1.54. Passes to the slot are highly dangerous, with an estimated 1.08. Rush passes meaningfully increase danger. Seam passes add moderate danger. Slot shots barely above the baseline. Danger drops at x extremes. All action type coefficients show excellent hairy caterpillar mixing. Both of the chains overlap randomly throughout all 500 post-warmup draws. Histograms are smooth and unimodal. No drifting, no sticking, no divergence is present. Convergence is clean across all action-type parameters. The posterior predictive check has the model-replicated data sitting almost perfectly on top of the observed data. The two lines are essentially indistinguishable. The U-shaped distribution reflects the binary 0/1 outcome. The model reproduces this shape, confirming that the Bernoulli likelihood is correctly specified and that the model is well calibrated to the observed danger rate of 21.9%.

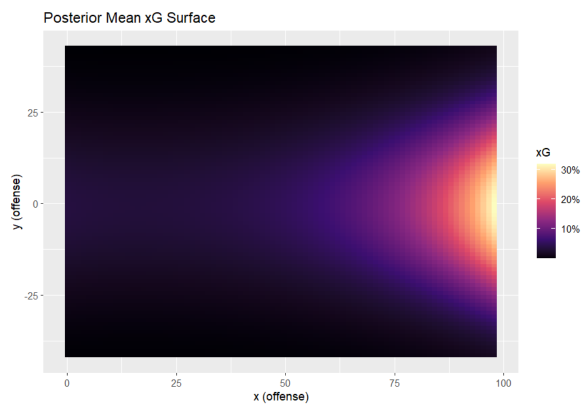
3. RESULTS

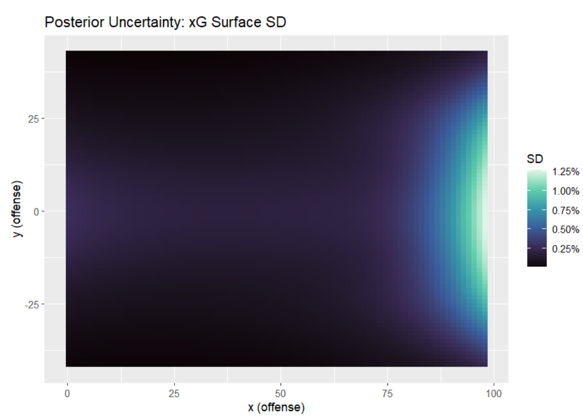
3.1. Question 1 Results

Question 1: How does xG vary by shot location and player?

The coefficients showed that shots closer to the center of the ice horizontally increase goal probability. Distance on the ice up and down from the goal does not matter on its own. The quadratic depth term was the strongest predictor of goal probability in the model. This suggests that shooting percentage does not decline linearly with distance from the net but accelerates, as shots beyond a certain depth threshold incur a compounding penalty on goal probability. This is consistent with the intuition that shots from the slot and crease areas are taken at dramatically higher rates than other types of shots. It is also important to note that the player's random effect term is very small once location is taken into consideration. Location accounts for much of the variance, while player identity adds very little. Then I compared the xG produced by my model with the xG variable in the dataset, listed as sl_xg_all_shots, to demonstrate a good fit. A correlation of 0.56 was calculated between my model and the AVS analytics team. This demonstrates that, with only location and player, roughly 32% of the variance is explained before adding any contextual features. Then the model was calibrated, and it was found that my model mean for xG was 0.499, compared to the AVS xG of 0.481. Nearly identical average xG values, calculated independently, strongly indicating that the model is well-specified.

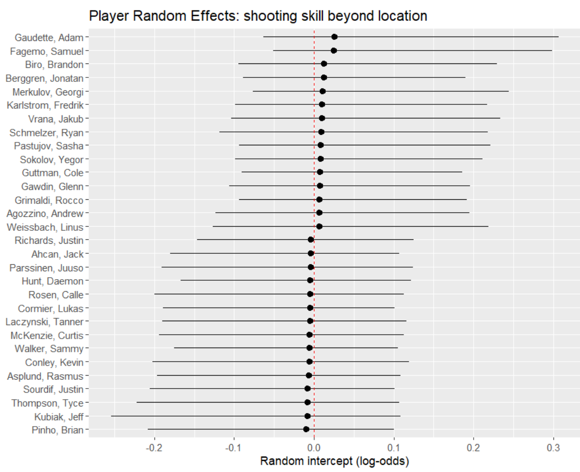
Next, heatmaps were created depicting the posterior mean xG and the posterior uncertainty in xG across the ice rink. The first heatmap is an 80x80 grid covering the offensive zone, with each tile colored by the model's posterior mean goal probability. This heatmap is based only on location effects, as player effects were deemed a trivial differentiator. The hot zone (yellow/orange, 30%+ xG) sits directly in front of the net at high x values centered on y equals zero, fanning outward in a cone shape. xG decays sharply with distance and angle. The corners and perimeter remain near 10%. This confirms that the model learned the correct hockey geography. Straight-on and close-range shots are dramatically more dangerous than shots from distance or wide angles. The second heatmap is on the same 80x80 offensive grid, colored by the standard deviation of the posterior draws per cell rather than the mean. This shows where the model is most confident versus where it is most uncertain in its xG estimates. Uncertainty is highest in the same high-danger zone near the net (standard deviation up to 1.25%), and near zero everywhere else. This is the expected pattern as the model is most uncertain precisely where xG is highest because that region has the most variance in its outcomes. The dark perimeter indicates the model is very confident that those locations are low-risk.





The next plot dives into player random effects. The top fifteen overperformers and bottom fifteen underperformers are extracted from the models' random intercepts, shown as point estimates with 95% credible intervals.

Positive values indicate a player scores more than their shot locations predict. While negative values indicate they underperform their locations. The red dashed line at zero is the league average after controlling for location.

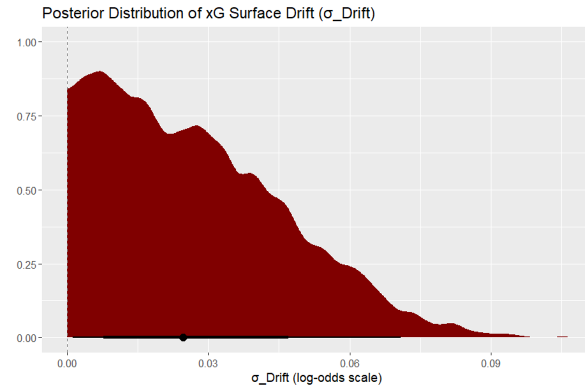


Three key findings emerge from this plot. First, all credible intervals are wide and almost all cross zero. Thus, the model shows that at the AHL level, shot volumes and individual player finishing skill cannot be distinguished from just plain noise. Second, all point estimates cluster within plus or minus 0.05 log-odds of zero, meaning the practical magnitude of player effects is tiny. Third, Adam Gaudette leads as the top overperformer, but his wide interval does reflect a limited shot volume. The overarching conclusion is that where you shoot from matters far more than who is shooting. Therefore, location explains the vast majority of goal probability, with individual finishing skill adding minimal additional insight.

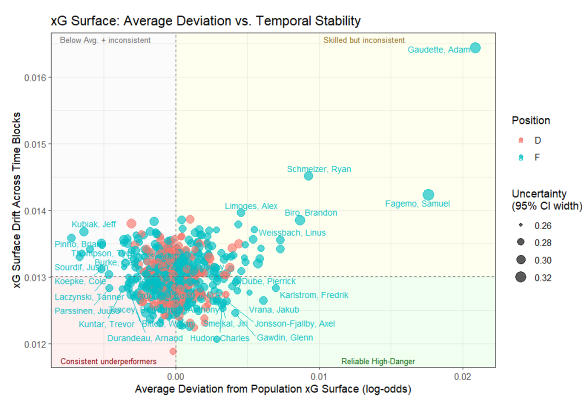
3.2. Question 2 Results

Question 2: Is that xG relationship stable across time, or does a player's shooting probability from a given location drift?

Sigma drift estimate is 0.03 with a 95% confidence interval from 0.00, 0.07. The upper bound is well below any practically meaningful threshold. The static xG surface for part one is confirmed to be a reliable seasonal indicator. Player shooting profiles do not meaningfully drift across a 10-game window.



The distribution plot is highly concentrated near zero with a long right tail. The bulk of the posterior mass lies between 0 and 0.03, indicating the model is highly confident that drift is small. The black median point estimate sits right at 0.0246, with the credible interval bar barely visible. The estimated temporal drift corresponds to a shift of just 0.10% in goal probability per 10-game window. This means part one's static xG surface is a trustworthy seasonal summary. A player's shooting profile from a given location does not meaningfully change across 10-game windows. Coaches and analysts can rely on single-season location-adjusted xG estimates without worrying that early-season shooting patterns will become obsolete by playoff time.



In this scatterplot, each point represents a player, positioned by how much they deviate from the population xG surface on average (x-axis) and by the drift of their shooting profile across 10 game windows (y-axis). Point size reflects uncertainty in the estimation, with larger points indicating players with fewer shots and therefore less reliable estimates. The dashed lines divide the plot into four quadrants, each with a distinct interpretation in hockey. The bottom-right is considered the reliable high-danger zone depicted in green. The players who consistently shoot from above-average locations with stable profiles across the season. Fagemo, Samuel, Gawdin, and Glenn are the clearest examples, as their positive x-axis deviation is sustained and their drift is below the median, meaning the part one xG surface accurately characterizes their entire season the most. The top-right is considered the skilled but inconsistent section, depicted in yellow. The players who shoot from high-danger locations on average, but whose spatial profiles shift meaningfully across blocks. Gaudette, Adam, is the most extreme outlier in the entire dataset, sitting far to the right and well above the median drift line. Notably, his point is large, indicating high uncertainty in estimation. Schmelzer, Ryan, and Limoges, Alex, also appear here. The bottom-left consistent is considered the underperformers depicted in red. The players who consistently shoot from below-average locations with stable profiles. Durandeau, Arnaud, and Kuntar, Trevor fall here, meaning their low xG is a reliable seasonal characteristic rather than a temporary slump. The top-left, below-average, and inconsistent players are depicted in gray. These players are shooting from poor locations with volatile profiles. Kubiak, Jeff, and Pinho, Brian appear here, suggesting their shooting profiles are both below average and unreliable across windows.

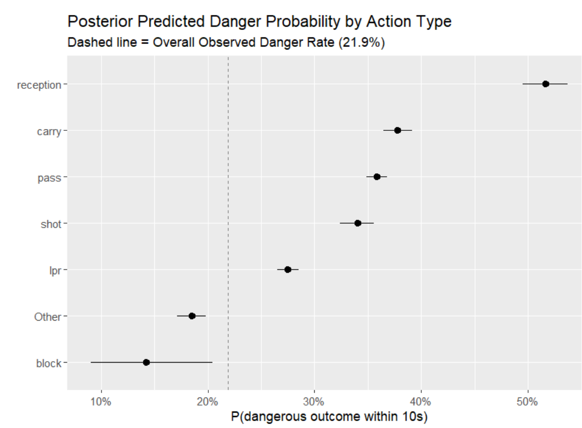
Key takeaways from this graphic are as follows. Most players are clustered closely around the plot's center. The y-axis range is narrow, spanning only 0.012 to 0.017. This extremely compressed range shows that temporal drift is negligible for almost all players. It also visually supports the sigma_drift = 0.025 result from the posterior distribution. Forwards (teal) and defensemen (pink) show no position-based separation in either dimension. Both positions mix throughout all quadrants, suggesting xG surface stability is a general property, not position-specific. Thirdly, the largest points indicate the most uncertainty in the estimates. The largest estimate appears on the extreme x-axis values. This supports partial pooling. Players far from the center mostly have fewer shots and wider credible intervals. The Bayesian hierarchical approach adds value by drawing extreme-looking results from low-volume players toward the population mean.

3.3. Question 3 Results

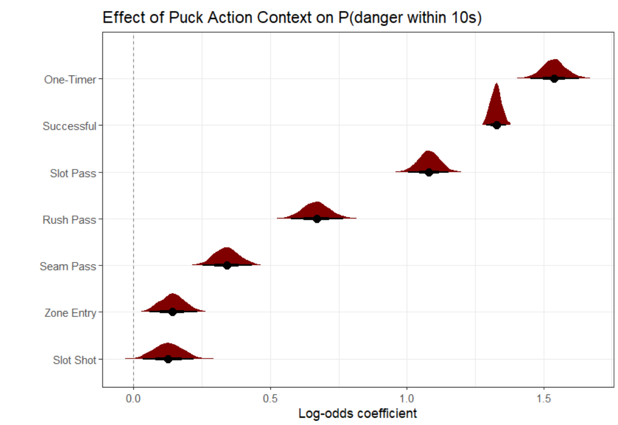
Question 3: How does each puck action change the probability of generating a dangerous offensive outcome (e.g., shot or slot shot) within the next 10 seconds, given spatial and game context?

A one-timer is defined as a shot taken in one motion, without stopping the puck first. A one-timer from anywhere (+1.54) is more predictive of danger than being in the slot with no special action context.

The above spatial heatmap shows the probability of a dangerous outcome within 10 seconds across all offensive zone locations, holding action type fixed with no special context. Unlike part one's xG surface, which peaked right in front of the net. This surface peaks in the middle of the offensive zone, around x equals 50, and decays toward the corners and the net. This is a different spatial pattern from part one. The actions in the middle of the offensive zone most frequently precede shots within 10 seconds because they are transition points where the puck moves from one player to another before a shot attempt. The y-axis decay confirms that actions along the boards are less likely to generate immediate danger than actions through the middle of the ice. Location alone explains only a narrow range of danger probability. Where you act matters less than how you act here.

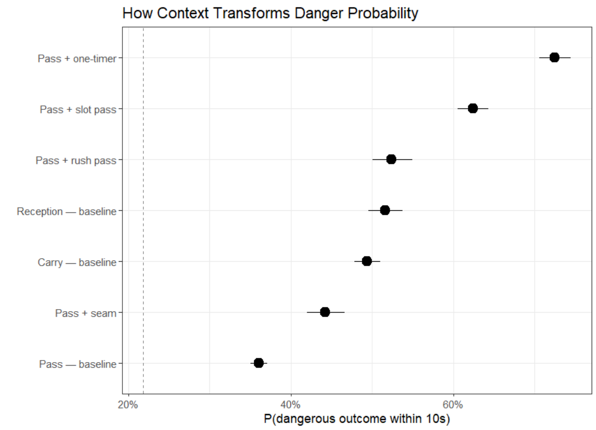


The plot above shows the posterior-estimated danger probability for each action type at the mean location, with no special context. The dashed line marks the overall observed danger rate of 21.9%. A reception creates the highest danger action by far (52%), nearly double the overall rate. Receiving a pass means you just gained possession in a position to shoot immediately, thus the 10-second window almost always captures the subsequent shot attempt. A carry, when a player is driving the puck toward the net, is the second most dangerous action, consistent with zone entry and rush situations being high-danger at 38%. A pass is the reference category, which sits well above the overall rate, reflecting that passes in the offensive zone frequently set up immediate shots at 36%. A shot itself poses danger within 10 seconds, primarily through rebounds at 33%. A shot that doesn't score often leads to another shot attempt. LPR, also known as loose puck recovery, generates moderate danger (27%), reflecting that regaining possession creates an offensive opportunity, but from a less controlled position than a reception. Actions that maintain possession and create shooting opportunities (reception, carry, pass) generate far more danger than actions that disrupt play or represent scramble situations.



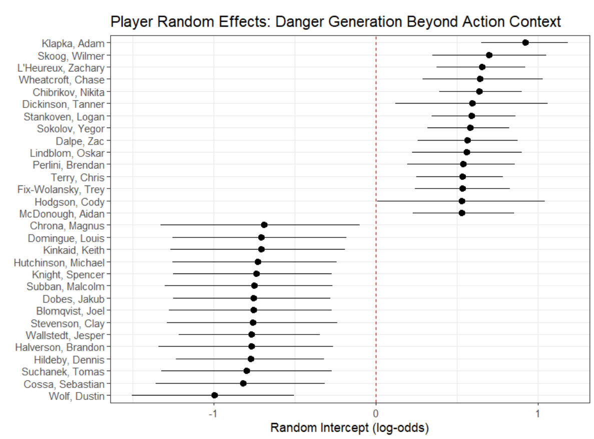
The plot above shows the posterior distributions of log-odds coefficients for each contextual binary feature, showing how much each feature shifts danger probability beyond what action type and location alone predict. All features shown are positive; every feature increases the danger. The one-timer log odds coefficient is 1.54. It is the dominant contextual predictor with the tightest, most certain posterior.

A one-timer increases the log-odds of danger by 1.54. This represents a roughly 15–20 percentage-point increase in the probability of danger. The narrow posterior distribution confirms this is estimated with high certainty from thousands of one-timer events. The action success is the second strongest predictor, with an extremely tight posterior given the massive sample size. Whether the action succeeds is nearly as predictive as whether it's a one-timer. The slot pass is when passing to the 'home plate' area creates danger comparable to a one-timer, reflecting that slot passes directly set up high-quality shot attempts (1.08). A rush pass that passes off the rush creates substantially more danger than a normal pass, capturing the transition game advantage (+0.67). It shows that context transforms the danger profile of an action more than the action type itself. A pass with a one-timer context is more dangerous than a reception without context.



This plot compares specific action-context combinations on the probability scale. A pass-and-one-timer (65%) is the most dangerous scenario in the dataset, nearly triple the 21.9% overall average danger rate. Setting up a one-timer with a pass is the most reliable way to generate a dangerous outcome. A pass-and-slot pass (60%) is nearly as dangerous, confirming puck movement into the home area is crucial. A pass-and-rush pass (48%) is far more dangerous than a static zone pass, demonstrating the transition advantage. Without special context, receiving a pass baseline (46%) or carrying the puck baseline (45%) generates more than double the overall danger rate. Seam passes add 40% more danger above baseline passes. Even a simple offensive-zone pass has a 33% probability, well above the 21.9% overall rate.

The gap between the pass baseline (33%) and the pass-and-one-timer combination (65%) is a 32-percentage-point difference, indicating the added value of a single contextual feature. It demonstrates that the same action can have radically different implications for danger depending on context.



This plot demonstrates individual player random intercepts after controlling for all action types, location, and contextual features. Players who consistently generate more or less danger than their actions and locations predict. The x-axis spans from -1.2 to +1.0 log-odds, a substantial range. Top overperformers are Klapka, Adam, L'Heureux, Zachary, Wheatcroft, Chase, and Chibri, Nikita, all of whom show large positive intercepts with intervals that confidently exclude zero, meaning they reliably generate danger beyond what their action types and locations predict. Wolf, Dustin, Cossette, Sebastian, and Suchanek, Tomas are the bottom underperformers, showing the largest negative effects, consistently generating less danger than expected. Most players exhibit substantial uncertainty, which correctly reflects limited per-player event counts even in a large dataset. This is a fundamentally different finding from parts one and two. In part one, the player's finishing skill was negligible after controlling for location. Where you shoot matters far more than who shoots. In part three, individual player skill in generating dangerous situations is substantial and real after controlling for action type and context. Some players are systematically better at creating danger regardless of what action they take or where they take it.

Overall, puck action type, success, and contextual features collectively explain far more variance in danger probability than shot location alone. Reception and carry are the most dangerous action types, but context dominates action type. A pass that sets up a one-timer (65% danger) is more dangerous than any action type without context. One-timers (+1.54 log-odds), action success (+1.33), and slot passes (+1.08) are the strongest individual predictors. Furthermore, individual players differ substantially in their ability to generate danger beyond what actions and locations predict (sd = 0.37), a skill completely invisible to traditional xG models that focus only on shot location.

4. DISCUSSION

4.1 Domain Significance

This study provides actionable insights for hockey operations at every level.

For coaches, the most practically significant result is the outsized impact of contextual action features on the probability of danger. Designing offensive systems that generate slot passes and one-timer opportunities is quantifiably superior. A one-timer context alone adds 32 percentage points of danger probability to a pass, meaning even a moderate increase in how frequently a team creates these situations can dramatically shift their offensive outcomes across a season.

For player evaluation, this study reveals a key distinction. Finishing skill and danger-creation skill are distinct and measurable abilities that traditional metrics combine into a single number. Location-adjusted xG from question one distinctly measures a player's ability to convert chances from high-quality locations, while the player random effects from question three specifically capture a player's ability to generate dangerous situations, independent of action or location. Thus, using both metrics together explicitly distinguishes between converting quality chances and actively creating danger, giving a more principled and comprehensive view of offensive contribution.

For analytics, the stability from question two builds confidence. Single-season location-adjusted xG does not need mid-season recalibration. Hierarchical model intervals help evaluate low-volume players without overreacting to small samples.

4.2 Limitations

Some limitations of my work include violations of independence assumptions, subsampling, and the omission of an interaction term. Shots within a game are not independent. A full Markov game model would account for sequential dependencies. This is a key limitation in question one and question three. Also, 20% subsample preserves outcome rates but may still under-represent rare action combinations. A full dataset would yield tighter posteriors. Lastly, no xy interaction term is present in question one. No action type by location interactions in question three. These could improve the spatial models.

4.3 Recommendations

Future recommendations include developing a full Markov game framework, extending the analysis to defensive metrics, and implementing real-time expected possession value. First, model the full sequence of puck actions as a Markov decision process to remove the independence assumption and capture how plays chain together. Next, extend the analysis to defensive metrics to complement current offensive metrics. Develop a parallel model for defensive disruptions, such as blocks and interceptions, to create a complete possession-value framework. Finally, use the current model as a foundation for real-time expected possession value by updating predictions after each action during live games to support bench-side decision-making.

4.4 Conclusions

First, shot location is the lead driver of expected goals, and individual finishing skill is largely insignificant. Where you shoot from matters more than who is shooting. Second, the static xG surface from question one is a reliable seasonal summary. Player shooting profiles do not drift across 10-game windows. Third, and most importantly, when the lens shifts from shot-making to danger creation, context reigns above everything else. The same action can have drastically different implications for danger depending on its context. A pass that sets up a one-timer is nearly triple the danger of a baseline pass. Individual players differ substantially in their ability to generate danger beyond what their actions and locations predict, a skill completely invisible to traditional xG models.

In sum, these findings show that the Bayesian hierarchical modeling gives hockey analytics a reliable new path. It quantifies uncertainty honestly, handles sparse player-level data responsibly, and reveals the contextual importance behind real offensive value.

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